

A Discrete-Component Second-Order CT $\Delta\Sigma$ ADC Design

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1 Overview

In the sigma-delta ($\Delta\Sigma$) ADC topology, a quantizer produces a stream of coarse 1-bit decisions, each of which is fed back through a 1-bit DAC into the loop’s cascaded integrators, shaping the input to subsequent decisions. The block diagram of the $\Delta\Sigma$ ADC is shown in figure 1.

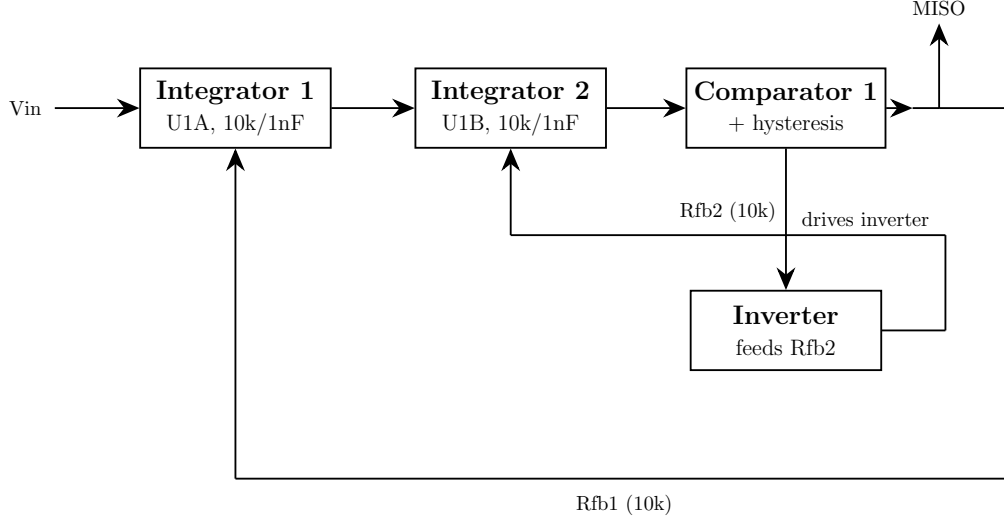


Figure 1: Second-order CIFB $\Delta\Sigma$ modulator, block-level view. Comparator 1’s decision is fed back through two separately-signed paths: directly via Rfb1 to Integrator 1, and through an inverting stage (Inverter) via Rfb2 to Integrator 2 — the sign correction discussed in §4.

The ones density of the bitstream, denoting the proportion of quantizer decisions resolved to logic 1 within a given interval, is a monotonic function of the input voltage centered around the comparator’s reference voltage, V_{ref} . Because resolution is built up from many decisions rather than a single high-precision comparison, a $\Delta\Sigma$ ADC trades conversion latency for resolution relative to a SAR ADC, which resolves a full-precision word in a single conversion cycle. This report documents a second-order continuous-time $\Delta\Sigma$ ADC, designed and tested in LTspice before discrete breadboard implementation using physical components. The target specifications for the design are listed in table 1.

Table 1: Design Specifications

Parameter	Value
Signal Bandwidth	200 Hz
Oversampling Ratio (OSR)	128
Sample Rate	51.2 kHz

The cascade-of-integrators feedback (CIFB) topology is accomplished using the dual op-amp MCP6002, followed by the LM393 comparator acting as the 1-bit quantizer, with

the quantizer's decision fed back through two separately-signed 1-bit DAC paths, one direct and one inverted, to correctly close negative feedback around each integrator stage. The system has been fully brought-up and its closed-loop dynamics are verified and stable. A five-point input sweep from 1.23 V to 3.26 V correctly demonstrated a monotonic trend between effective ones density and input voltage without latching to either rail.

Achieving this correct behavior is the subject of this report. This report documents the debugging process through which the $\Delta\Sigma$ ADC design was built on physical hardware, especially diagnosing and fixing a sign error which mistakenly caused positive feedback, and several instrumentation artifacts that initially presented as hardware faults. One known limitation remains open: occasional long runs of identical bits (limit cycles) near mid-scale, an expected consequence of the unlocked, free-running quantizer, with a known synchronous fix using the 74HC74 identified but deferred due to scope.

2 Architecture

Figure 2 shows the circuit-level schematic of the $\Delta\Sigma$ ADC.

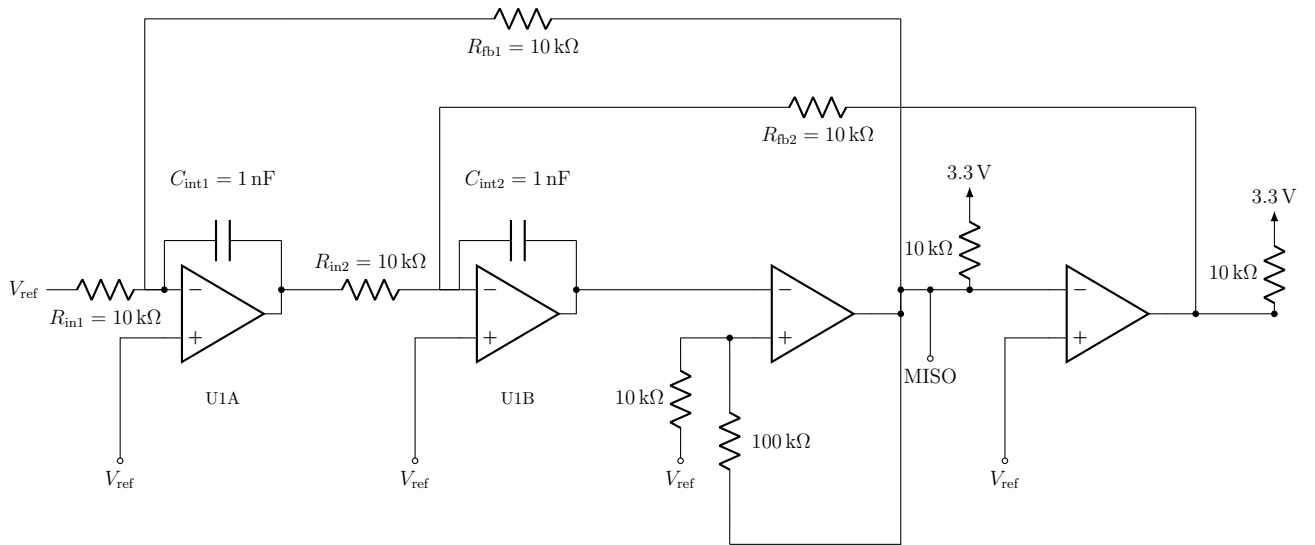


Figure 2: Full discrete-component schematic. U1A and U1B are the two amplifiers of a single MCP6002; Comparator 1 and Comparator 2 are the two comparators of a single LM393, with Comparator 2 wired as a unity inverter feeding Rfb2. Component values as built; final values summarized in Table ??.

3 Bring-Up Methodology

3.1 Root Cause: DAC Sign Error

3.2 Debugging Highlight: Instrumentation artifacts

4 Results

5 Limitations

6 Next Steps

References

A Source Code and Reproducibility